

Project Design

1. Problem definition

Organizations frequently import data into ServiceNow from external files like Excel or CSVs. At present, this workflow is largely manual, requiring users to upload the files, map fields, and establish connections between different tables by hand.

This manual approach introduces several challenges. It is highly time-consuming and demands repetitive labor. Additionally, manual field mapping is prone to human error, which often leads to inaccurate or incomplete records. Managing table relationships—such as connecting users to their respective departments or managers—is also complex and prone to confusion. Consequently, users spend valuable time verifying and fixing data, which ultimately lowers productivity.

Solution Definition

To solve these problems, an automated approach was created using native platform features like Data Sources, Import Sets, Transform Maps, and Dot Walking. This system lets users easily upload files through a Data Source. The information is then held in temporary staging tables called Import Sets. Next, Transform Maps automatically align source fields to target fields to reduce manual work. To finish, Dot Walking creates table relationships by reading data through reference fields. Overall, this solution speeds up data handling, cuts down on mistakes, improves reliability, and makes the whole import and mapping process much simpler.

2. Solution Architecture

The architecture shows how the system is put together and how its parts collaborate. The solution utilizes a layered workflow:

- **Input Layer:** Takes in external files like CSVs or Excel spreadsheets. Users push this data into the system via the Data Source tool.
- **Processing Layer:** Features Import Sets and Transform Maps. Import Sets hold the incoming data temporarily. Transform Maps then translate those fields to the final table, with optional scripts available to handle **extra automation**.
- **Relationship Layer:** A major part of this project, this tier uses Dot Walking and reference links to join tables together. It can natively attach user files to their managers or departments without manual assistance.
- **Data Layer:** Includes the live tables where the successfully handled data is saved, keeping system directories organized and well-maintained.
- **Output Layer:** Offers transaction logs and summary results. This step proves whether the import was successful and flags any errors that surfaced along the way.

Flowchart

